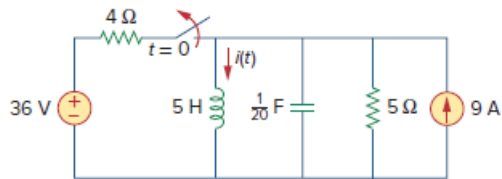


Problem

Determine $i(t)$ for $t > 0$ in the circuit of Fig. 16.66.

Figure 16.66:



Step-by-step solution

Step 1 of 8

(Refer Figure 16.66 from textbook) Consider the switch is close for time $t < 0$. Then, replace the inductor by short circuit and the capacitor by open circuit.

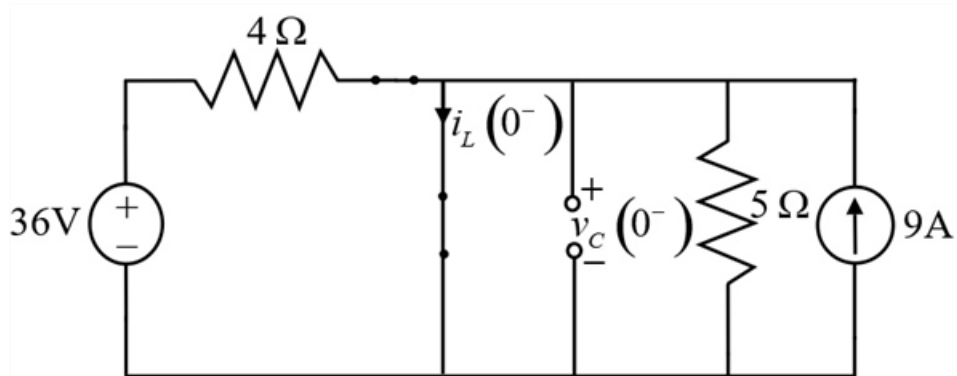


Figure 1

Step 2 of 8

Calculate the current $i_L(0^-)$ across the inductor.

$$\begin{aligned} i_L(0^-) &= \frac{36}{4} + 9 \\ &= 18 \text{ A} \end{aligned}$$

Calculate the voltage across the capacitor.

$$v_C(0^-) = 0 \text{ V}$$

Step 3 of 8

Consider the switch is open for time $t > 0$. Then, represent the circuit in s-domain.

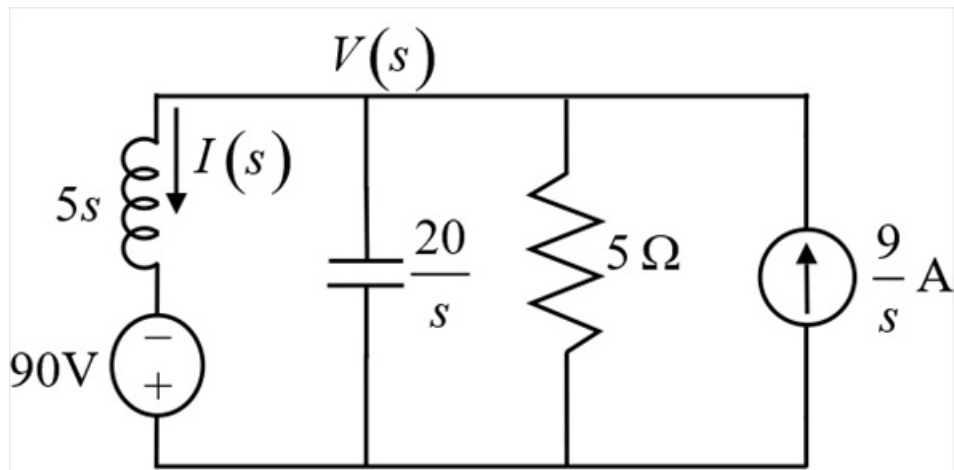


Figure 2

Step 4 of 8

Apply the nodal analysis in Figure 2.

$$\frac{V(s) + 90}{5s} + \frac{V(s)}{\left(\frac{20}{s}\right)} + \frac{V(s)}{5} = \frac{9}{s}$$

$$\frac{4V(s) + 360 + s^2V(s) + 4sV(s)}{20s} = \frac{9}{s}$$

$$s^2V(s) + 4sV(s) + 4V(s) = 180 - 360$$

$$V(s) = \frac{-180}{s^2 + 4s + 4}$$

Rewrite the above equation.

$$V(s) = \frac{-180}{(s+2)^2}$$

Step 5 of 8

The current $I(s)$ across the inductor is,

$$I(s) = \frac{V(s) + 90}{5s}$$

Substitute $\frac{-180}{(s+2)^2}$ for $V(s)$.

$$\begin{aligned} I(s) &= \frac{\left(\frac{-180}{(s+2)^2}\right) + 90}{5s} \\ &= \frac{-180 + 90(s^2 + 4s + 4)}{5s(s+2)^2} \\ &= \frac{-36 + 18s^2 + 72 + 72s}{s(s+2)^2} \end{aligned}$$

$$= \frac{18s^2 + 72s + 36}{s(s+2)^2} \dots\dots (1)$$

Step 6 of 8

Do the partial fraction of equation (1).

$$\frac{18s^2 + 72s + 36}{s(s+2)^2} = \frac{A}{s} + \frac{B}{(s+2)} + \frac{C}{(s+2)^2} \dots\dots (2)$$

$$18s^2 + 72s + 36 = A(s+2)^2 + B(s)(s+2) + C(s)$$

$$18s^2 + 72s + 36 = s^2(A+B) + s(4A+2B+C) + 4A$$

Compare the coefficients of s^2 , s^1 and s^0 .

$$A = 9$$

$$B = 9$$

$$C = 18$$

Step 7 of 8

Substitute 9 for A , 9 for B and 18 for C in equation (2).

$$\frac{18s^2 + 72s + 36}{s(s+2)^2} = \frac{9}{s} + \frac{9}{(s+2)} + \frac{18}{(s+2)^2}$$

Rewrite the equation.

$$I(s) = \frac{9}{s} + \frac{9}{(s+2)} + \frac{18}{(s+2)^2} \dots\dots (3)$$

Step 8 of 8

The inverse Laplace transform of $\frac{1}{s^2}$ is,

$$\frac{1}{s^2} \leftrightarrow t$$

The shifting property of inverse Laplace transform is,

$$\frac{1}{s+a} \leftrightarrow e^{-at}$$

Take the inverse Laplace transform of equation (3).

$$i(t) = 9 + 9e^{-2t} + 18te^{-2t}$$

Hence, the current across the inductor is $\boxed{9 + 9e^{-2t} + 18te^{-2t} \text{ A}}$.